

Detailed DSA Case Study - Array Problems

Case Study 1 - Smart City Traffic Analysis

Real-Time Scenario

A smart city installs AI traffic cameras at a busy junction. Every hour the camera stores the number of vehicles that crossed the signal. The traffic department wants to analyse the data to identify rush hours and improve signal timing.

Hour	6AM	7AM	8AM	9AM	10AM	11AM
Vehicles	120	250	430	300	180	90

Visual Representation

- Data is stored in arrays.
- Traverse from first element to last.
- Compare each value with current Max and Min.
- Add every value to Total.
- Average = Total / Number of Elements.

Interview Tasks

- Find the peak traffic hour.
- Find the lowest traffic hour.
- Calculate total vehicles.
- Calculate average traffic.
- Count hours with traffic greater than 250.
- Display the busiest and least busy hours.

Pseudo Algorithm

1. Store hours and vehicle counts in arrays.
2. Initialize max, min and total using the first value.
3. Traverse the array once.
4. Update max and min whenever required.
5. Add every value to total.
6. Compute average = total / length.

Dry Run

Start with max=120, min=120. At 8AM, max becomes 430. At 11AM, min becomes 90. Total becomes 1370. Average = $1370/6 = 228.33$.

Expected Output

Peak Hour = 8AM (430) Lowest Hour = 11AM (90) Total Vehicles = 1370 Average = 228.33

Case Study 2 - Shopping Mall Weekly Sales

Real-Time Scenario

A shopping mall stores the revenue generated every day. The manager wants to analyse the week's performance before planning discounts and staff allocation.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Sales	4500	3200	7800	6100	9000	12000	8500

Visual Representation

- Data is stored in arrays.
- Traverse from first element to last.
- Compare each value with current Max and Min.
- Add every value to Total.
- Average = Total / Number of Elements.

Interview Tasks

- Find the best sales day.
- Find the worst sales day.
- Find the Top 3 sales.
- Calculate total revenue.
- Calculate average daily sales.
- Count days where sales exceeded ■8000.

Pseudo Algorithm

1. Store day names and sales in arrays.
2. Traverse the sales array.
3. Track maximum, minimum and total.
4. Maintain first, second and third largest values for Top 3.
5. Print final report.

Dry Run

Highest sale becomes ■12000 on Saturday. Lowest becomes ■3200 on Tuesday. Total revenue = ■51100. Top 3 = ■12000, ■9000, ■8500.

Expected Output

Best Day = Saturday Worst Day = Tuesday Top 3 = 12000, 9000, 8500 Total Revenue = ■51100